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INDUSTRY EXPERTISE

- C
- Objective-C
- Swift
- C++
- JavaScript
- C#
- LUA
- Python
- Ruby

STRENGTHS

- Logical approach to problem-solving
- Initiative: ability to learn new things on your own accord
- Communication skills: can easily demystify technical jargon
- Confident at training internal & external stakeholders
- Highly analytical: can develop solutions to complex problems

TOOLS & PLATFORMS

- AI/LLM Tooling (MCPs, Skills, Plugins, Agents)
- GitHub Actions / CI/CD
- Datadog (RUM, Dashboards, APM)
- Jira Automation
- n8n (Workflow Automation)

JOSEPH MATTIELLO

Mobile Architect & Engineer

SUMMARY

Engineering Leader and Mobile Architect, 17+ years iOS — started before the SDK shipped. Creator of Provenance EMU (App Store, 388k+ downloads, 6.3k stars). Three US patents (Sense Networks). Previously Director of iOS Platform at NewsCorp, leading an 8-region global team. Currently on Wayfair's App Infrastructure team — ~$2M/year in bandwidth savings, custom MCP servers and AI tooling for 60+ mobile engineers. The same agent tooling shipped five Virtual Jaguar emulator releases in 24 days. Selectively considering Head of AI Platform / VP Engineering / Director / CTO roles plus a small number of consulting engagements. Details at joemattiello.dev.

EXPERIENCE

L4 Mobile Engineer — App Infrastructure

Wayfair

09/2025 - Present — New York, NY (Remote)

- Shipped MP4 compression and caching strategy producing ~$2M/year in recurring bandwidth savings.
- Built custom MCP servers, Claude skills/plugins, and n8n automation for 60+ mobile engineers.
- Designed bootup task performance tracking system with Datadog dashboards surfacing release-over-release regressions.
- Integrated Datadog RUM into the iOS app for real-time per-release visibility into user-experience metrics.

L4 Mobile Engineer — Product Discovery

Wayfair

01/2025 - 09/2025 — New York, NY (Remote)

- Led and contributed to the development and enhancement of the product discovery feature-set, improving user engagement and conversion.
- Spearheaded initiatives to optimize application startup performance and reduce network overhead, resulting in a faster and more responsive user experience.
- Revamped homepage layout and implemented modern UI elements, including shimmer effects for loading states, to improve visual appeal and perceived performance.
- Collaborated on the design and implementation of new UI components for upcoming features, ensuring a consistent and intuitive user interface.
- Proactively identified and addressed performance bottlenecks across multiple application modules and teams, driving significant improvements in overall app stability and speed.

Founder

Matti Media LLC

05/2021 - Present — New York, NY

- Creator of Provenance-EMU, a multi-platform video game console emulator for iOS/tvOS/macOS, built entirely in Swift. 100k+ downloads in first few months.
- Ported dozens of C/C++ emulators to Swift Package Manager, optimized for ARM platforms, and applied various performance and compatibility patches.

SDKs & APIs

- Core Audio
- Core MIDI
- iOS/tvOS SDK
- Realm
- SwiftUI
- UIKit
- XCTest
- AppKit
- AVFoundation
- Combine
- Core Animation
- Core Location / MapKit
- RxSwift
- StoreKit
- Max/MSP
- Metal
- OpenGL/ES
- React Native

EDUCATION

BS, Computer Science

University at Buffalo

PROJECTS

Provenance

tvOS & iOS multi-platform emulator. Author and PM.

Virtual Jaguar (libretro)

Maintainer of the hard-forked Atari Jaguar core and author of its desktop app for macOS, Windows and Linux (v3.9.0). The first cross-platform open-source Jaguar CD emulation and the first shipped Jaguar Voice Modem emulation; 1,559 commits since v2.3.2.

Hero Transitions

- Developed a full SwiftUI user interface and integrated native iOS features like Spotlight, QuickLook, FileProvider, Stickers, and Siri extensions.

Contract iOS Engineer

Fitness AI

04/2021 - 05/2021 — New York, NY

- Improved project organization and code structure.
- Implemented comprehensive crash and console logging with breadcrumbs for improved error tracking.
- Developed new UI features, including drag-and-drop reordering for table view items.
- Addressed performance bottlenecks and enhanced application stability.

Principal iOS Engineer

Cameo

10/2020 - 03/2021 — New York, NY

- Served as the first native iOS Engineer on a mature ReactNative stack, bridging native capabilities with cross-platform development.
- Integrated HD asynchronous video compression, significantly improving video quality while reducing bandwidth and AWS costs.
- Implemented AR filter support, enhancing visual quality and user experience based on feedback from top talent.
- Focused on optimizing performance and ensuring seamless feature integration.
- Reduced memory overhead, improved startup and login times, and optimized network usage.
- Migrated and consolidated CI/CD pipelines from shell scripts/CircleCI to GitHub Actions, achieving over 10x speed improvements in PR validation.

Director iOS Platform (Global HQ)

NewsCorp Inc.

04/2019 - 02/2020 — New York, NY

- As iOS Platform Lead in the central NewsCorp app platform team (one of 8 core members), directed and managed internal, sub-brand, and contract developers working on platform and brand-specific applications.
- Managed iOS developers across multiple global regions (NYC, Argentina, London, Ukraine, Russia, Hong Kong, Sydney & Bangalore) working in parallel on various apps.
- Led the team that built the core SDK foundation used by all NewsCorp brands (e.g., DowJones, NYPost, The Australian, Barrons, Times UK) for their mobile applications. Directly managed platform SDK codebase, pull requests, and internal engineering resources loaned to brand app teams.
- Fostered a private "open source" model for SDKs, enabling internal and brand developers to contribute feature requests and pull requests, with rapid integration via platform town halls.
- Responsibilities included establishing CI/CD for brand apps, creating documentation, tutorials, code samples, and app templates for developers.
- Established and staffed a new full-time iOS satellite team in Barcelona; also responsible for restaffing and training existing teams in Argentina (Spark), Bangalore (NTS), and Ukraine/Russia (EPAM).
- Hired, trained, and led the iOS team for the new Knewz app, a high-profile project overseen by CEO Robert Thomson and Chairman Rupert Murdoch. Secured an unsolicited App Store promotion from Apple. Maintained a 4.5+ star rating during tenure.

Lead Developer

Salido Inc.

Maintained and contributed to Hero, a library for building custom view...

RxReachability

Took over maintenance for RxReachability, a RxSwift wrapper for Apple's...

Bass Station 2 for Max/MSP

Developed a Max for Live device to control the Novation Bass Station II synthesizer.

iPhreak

Created a native DTMF generator for the original iPhone (Objective-C/GPL)...

Question of the Day

Contracted to add features to an existing iPhone application, "Question of the Day."

App Store

MixMan DM2 USB/MIDI driver for OS X

(unofficial, reverse engineered protocol) Home, MacUpdate Reviews (C and C++...

OS X driver for Starr Labs USB Guitar

Developed an OS X USB driver for Starr Labs MIDI guitars (Objective-C & C).

OS X driver for Aurora Mixer

Created an OS X driver for the Aurora digital audio mixer (Objective-C & C),...

Pbxbuild

Submitted major patches to the Pbxbuild tool, enabling conversion of complex...

OPEN SOURCE

UTM

Contributed a pull request to UTM, a popular virtual machine host for iOS and macOS.

Project Link PR #3055

retroarch

Wrote the AVFoundation camera driver for iOS/macOS/tvOS, credited by name in...

Project Link

PR #17657 (AVFoundation camera driver credit)

PR #17569 (CoreMIDI driver)

PR #17591 (CoreLocation driver)

PR #17590 (CoreAudio microphone driver)

PR #17990 (macOS CoreAudio microphone split)

reicast

Contributed UIKit UI to reicast, a popular emulator for the Sega Dreamcast.

Project Link Flycast (Successor)

RxRealm (RxSwiftCommunity)

Maintainer and contributor to RxRealm, providing RxSwift bindings for Realm...

08/2018 - 04/2019 New York, NY

- As Lead Developer, managed the front-end development team, coordinated with back-end and other front-end teams on feature development, and led architectural design for new technical business relationships and 3rd-party vendor integrations.
- Developed the first Swift-native XML-SOAP framework with full Codable support while integrating with legacy vendors. Refactored the monolithic project into independent frameworks and separate repositories, streamlining global team coordination using CI/CD tools (CircleCI, Fastlane, Carthage, Rome, custom scripts) to mitigate QA and merge conflicts.
- Actively created development tools, documentation, and templates to promote improved organization and coding standards. Introduced the team to modern development practices including Promises, RxSwift, MVVM-C, Clean Architecture, and Objective-C to Swift interoperability.

Lead Mobile Architect / Team Lead

Hearst Digital News

05/2012 - 06/2018 New York, NY

- As Lead Developer, responsibilities included code review, peer training, ticket delegation, build submission, and release scheduling in collaboration with project managers. Managed a team of dedicated mobile developers with occasional cross-development with web and back-end teams.
- Maintained a 4+ star App Store rating through multiple update releases. Initiated conversion of the codebase to Objective-C 2.0 and ARC, resulting in an 80%+ reduction in crash reports and a 30% reduction in memory footprint. Refactored the shared codebase across four apps into a tiered build system of shared base and UI libraries, enabling future mobile/desktop app development and code sharing across Hearst teams.
- From Summer 2015, led the architecture and development of a new universal app (iOS 8+) using Swift 2.0, Objective-C, CoreData, CoreText, and Storyboards with size classes. Supported Apple Watch, Today Screen, and Force Touch extensions. Served as lead architect and developer, supported by two freelance developers. Example: San Antonio Express-News.
- Assumed role of network and security engineer, establishing office/colo infrastructure: intranet, firewalls, VPN, file/printer sharing, OS X Collaboration services (iCal, Address Book, WIKI). Built, tested, and deployed Hadoop systems for analytics.

Software Engineer (iOS/macOS)

Sooloos (Roonlabs)

05/2010 - 04/2012 New York, NY

- Linked existing C#/Mono audio decoders and playback functionality to native Core Audio, contributing significantly to the Novell MonoMac project for missing Cocoa frameworks.
- Developed native event and UI code for OS X for a cross-platform C#/OpenGL desktop application. Duties included native font rendering, window/menu management, mouse/keyboard events, native text entry, and process forking. Developed MonoMac patches and gained deep knowledge of NSApplication runloop, event handling, and OpenGL compositing.
- Developed an iOS app for two-way syncing and playback of Sooloos media libraries, interfacing with a cross-platform C toolkit for network communication and database management. Sole Objective-C coder and UI designer for this project.
- Overhauled the iPad remote control software, Core Control (formerly Sooloos), originally built in LUA with iPhone Wax. Involved extensive UI modifications, new layouts, expanded network protocol, and new functionality. Primarily LUA with performance-critical sections in Objective-C & C. Achieved 4.5+ star rating.

Software Engineer (iOS/DevOps)

Sense Networks, Inc.

Project Repository

XMLCoder

Contributed multiple significant pull requests to XMLCoder, a Swift library...

Project Link

PR #73 (Attributed Intrinsic Value Coding)

PR #70 (Dynamic Node Encoding & Fixes)

GLideN64

Contributed to GLideN64, an open-source graphics plugin for N64 emulators.

Project Link

SideJITServer

Contributed to SideJITServer, a project enabling JIT for sideloaded iOS...

Project Repository

Pull Request #150

Mono

Authored C# wrappers by hand for CoreMIDI, CoreAudio, and other Apple...

Project Link

MonoMac PR #20 (CoreAudio, CoreMIDI)

MonoMac PR #4 (AppKit, Foundation, etc.)

MultiPatch

Submitted a pull request (PR #19) to MultiPatch, a versatile ROM patching...

Project Repository PR #19

OSSSpeechKit

Added Swift Package Manager (SwiftPM) support to OSSSpeechKit, enabling...

Project Repository

PR #34 (SwiftPM Support)

sReto

Contributed to sReto, a P2P framework for real-time collaboration in Swift.

Project Link

UIForLumberjack

Contributed to UIForLumberjack, an iOS UI library for displaying...

Project Link

Play! Framework

Contributed a pull request to the Play! Framework, a high-velocity web...

Project Link PR #44

EGYWebViewController

Contributed to EGYWebViewController, an in-app browser component for iOS.

Project Link

Atari Jaguar 240p Test Suite

03/2008 - 03/2010 New York, NY

- Designed and developed Citysense for iPhone pre-official SDK, leveraging API hacking expertise. Managed full codebase and design. Named on patents for contributions to backend data analysis systems. Featured on GigaOM & NY Times.
- Assumed role of network and security engineer, establishing office/colo infrastructure: intranet, firewalls, VPN, file/printer sharing, OS X Collaboration services (iCal, Address Book, WIKI). Built, tested, and deployed Hadoop systems for analytics.
- As a software engineer, created utilities for managing large datasets and authored significant patches for GNUstep's PBXBuild tool (XCode to GNUstep Makefile conversion), becoming an official FSF/GNU project developer.
- Developed CabSense for iPhone, contributing to design and development with the Android team. Created pre-rendered assets and the Quartz 'Radar' view. Ensured secure server/client communication. Created reusable C/Cocoa libraries for third-party CabSense data interaction. Featured on VentureBeat, ZDNet, NY Times.

Security Consultant (Contract)

People Security

10/2007 - 12/2007 New York, NY

- Researched and developed training materials for an online "secure coding practices" school in direct partnership with Dr. Hugh Thompson.
- Covered topics such as avoiding "stack overflows," "nop sleds," "off-by-one" errors, other "buffer overflow" attacks, and advanced attacks against randomized stack/library addresses.

Associate Producer / Researcher

AT&T Tech Channel

08/2007 - 11/2007 Bedminster, NJ

- Researched and consulted on guests and topics for the AT&T Tech Channel. Managed bookings and discussion topics with prominent figures including Emanuel Goldstein (2600), Whit Diffie (RSA), Bill Cheswick (Firewalls), Jack Dorsey (Twitter), and H.D. Moore (Metasploit).
- Worked out of the main Bedminster, NJ, former Bell Labs research facility.

www.joemattiello.dev mail@joemattiello.com

Maintainer and contributor to the Jaguar 240p Test Suite for display...

Project Repository

PokeMini

Contributed to the PokeMini Libretro core, an emulator for the Pokémon Mini...

Project Link

PR #17 (Buffer Overflow Fix)

OBD2Connect

Added Swift Package Manager (SwiftPM) support, GitHub Actions validation,...

Project Repository

PR #4 (SwiftPM Support & Enhancements)

PINCache.Swift (Fork)

Maintained a fork of PINCache.Swift, a fast, non-deadlocking parallel object...

GitHub Repository